# 01NAEX - Lecture 04
# Latin Squares, Graeco-Latin Squares and Balanced Incomplete Block Designs

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# More Blocking Designs

## D. C. Montgomery, Design and Analysis of Experiments: Sections 4.2–4.4

**Last lecture:** the randomized complete block design (RCBD) removes the variability of **one** known and controllable nuisance factor from the experimental error.

**Today:** what if

- there are **two** nuisance factors (batch and operator, day and position)?
  $\Rightarrow$ **Latin square** design;
- there are **three** nuisance factors?
  $\Rightarrow$ **Graeco-Latin square** design;
- a block is too small to hold all treatments (a batch is enough for 3 runs only, a car can be tested with 4 additives only)?
  $\Rightarrow$ **balanced incomplete block design** (BIBD).

At the end: **Team Project 1** (pendulum experiment), in which you design, run and analyse your own blocked factorial experiment.

# The Latin Square Design

A $p \times p$ Latin square is a square of $p$ rows and $p$ columns; each of the $p^2$ cells contains one of $p$ letters (treatments), and **each letter occurs once and only once in each row and each column**:

| $4 \times 4$ | $5 \times 5$ | $6 \times 6$ |
|---|---|---|
| A B D C | A D B E C | A D C E B F |
| B C A D | D A C B E | B A E C F D |
| C D B A | C B E D A | C E D F A B |
| D A C B | B E A C D | D C F B E A |
| | E C D A B | F B A D C E |
| | | E F B A D C |

- Rows and columns are two blocking factors (two restrictions on randomization): the design blocks in two directions.
- Rows, columns and treatments are mutually orthogonal, so their sums of squares separate exactly.
- Only $p^2$ runs for three factors with $p$ levels each; the price is the assumption of **no interactions** between rows, columns and treatments.

A Sudoku grid is a $9 \times 9$ Latin square with an extra condition on the $3 \times 3$ boxes.

# Example: Rocket Propellant

## Montgomery, Example 4.3

Five formulations of a rocket propellant (A–E) are compared by their burning rate. Each batch of raw material is large enough for five formulations only, and the formulations are prepared by five operators who may differ in skill. Both batches and operators are nuisance factors: each formulation is tested exactly once in each batch and exactly once by each operator.

**Latin Square Design for the Rocket Propellant Problem**

| Batches of Raw Material | Operators | | | | |
|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 |
| 1 | $A = 24$ | $B = 20$ | $C = 19$ | $D = 24$ | $E = 24$ |
| 2 | $B = 17$ | $C = 24$ | $D = 30$ | $E = 27$ | $A = 36$ |
| 3 | $C = 18$ | $D = 38$ | $E = 26$ | $A = 27$ | $B = 21$ |
| 4 | $D = 26$ | $E = 31$ | $A = 26$ | $B = 23$ | $C = 22$ |
| 5 | $E = 22$ | $A = 30$ | $B = 20$ | $C = 29$ | $D = 31$ |

Rows = batches, columns = operators, letters = formulations; $N = p^2 = 25$ runs.

# Latin Square - Model and ANOVA

Effects model ($i$ = row, $j$ = treatment, $k$ = column; only two indices are needed to identify a cell):

$$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}, \qquad i, j, k = 1, \ldots, p, \qquad \varepsilon_{ijk} \sim N(0, \sigma^2),$$

$\sum \alpha_i = \sum \tau_j = \sum \beta_k = 0$, no interactions. $SS_T = SS_{Rows} + SS_{Columns} + SS_{Treatments} + SS_E$.

| Source | df | Sum of squares | $F_0$ |
|---|---|---|---|
| Treatments | $p-1$ | $\frac{1}{p}\sum_j y_{.j.}^2 - \frac{y_{...}^2}{N}$ | $MS_{Treatments}/MS_E$ |
| Rows | $p-1$ | $\frac{1}{p}\sum_i y_{i..}^2 - \frac{y_{...}^2}{N}$ | |
| Columns | $p-1$ | $\frac{1}{p}\sum_k y_{..k}^2 - \frac{y_{...}^2}{N}$ | |
| Error | $(p-2)(p-1)$ | by subtraction | |
| Total | $p^2-1$ | $\sum\sum\sum y_{ijk}^2 - \frac{y_{...}^2}{N}$ | |

Small squares leave few error df: $3 \times 3$ has 2, $4 \times 4$ has 6, $5 \times 5$ has 12. The analysis is a direct extension of the RCBD (Lecture 3).

# Constructing and Randomizing a Latin Square

**Standard Latin square:** first row and first column in alphabetical order (e.g. a cyclic square, each row shifted by one).

**Number of Latin squares** grows very fast:

| $p$ | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|
| standard squares | 1 | 4 | 56 | 9 408 | 16 942 080 |
| all squares | 12 | 576 | 161 280 | 812 851 200 | $\approx 6.1 \cdot 10^{13}$ |

**Randomization in practice:**

1. Choose a standard $p \times p$ square (ideally at random among the standard squares).
2. Randomly permute the rows, the columns and the treatment labels, independently. Orthogonality and balance are preserved.
3. Map rows and columns to the nuisance factors (e.g. row = day, column = operator); if a row is a day, the runs within the day are done in random order.

*Q:* Which mapping (row = day, column = operator or the reverse) is easier in your lab, and does it matter for the analysis?

# Rocket Propellant - ANOVA in Python

Data: `rocket.txt` (columns `y`, `batch`, `op`, `treat`).

```
URL = ("https://raw.githubusercontent.com/francji1/01NAEX/"
       "main/data/")
rocket = pd.read_csv(URL + "rocket.txt", sep=r"\s+")
ls = smf.ols("y ~ C(batch) + C(op) + C(treat)",
             data=rocket).fit()
sm.stats.anova_lm(ls, typ=2)
            sum_sq    df        F    PR(>F)
C(batch)      68.0   4.0   1.594    0.2391
C(op)        150.0   4.0   3.516    0.0404
C(treat)     330.0   4.0   7.734    0.0025
Residual     128.0  12.0
```

- The formulations differ significantly in mean burning rate ($F_0 = 7.73$, $p = 0.0025$).
- There is some evidence of differences between operators ($p = 0.040$).
- No strong evidence of differences between batches ($p = 0.24$). Blocking on batches is still a good precaution: we did not know it in advance.

# Rocket Propellant - Which Formulations Differ?

Tukey's HSD with $MS_E = 10.67$ and $df_E = 12$ of the Latin square model ($n = p = 5$ runs per formulation):

```
means = rocket.groupby("treat").y.mean()
# A 28.6   B 20.2   C 22.4   D 29.8   E 26.0
q = stats.studentized_range.ppf(0.95, 5, ls.df_resid)   # 4.508
hsd = q * np.sqrt(ls.mse_resid / 5)                     # 6.58
```

- $D - B = 9.6$, $A - B = 8.4$ and $D - C = 7.4$ exceed 6.58: formulations D and A burn faster than B, and D faster than C. The other differences are not significant.
- As in Lecture 3, `pairwise_tukeyhsd()` would fit a one-way model and put the operator and batch variability into the error.

**Ignoring the batches** (an RCBD with operators as blocks):

```
smf.ols("y ~ C(op) + C(treat)", data=rocket).fit()
# Residual SS 196.0 on 16 df, MS_E = 12.25, treat F = 6.73
```

Here the batch blocking cost 4 error df and reduced $MS_E$ only from 12.25 to 10.67.

# Replicated Latin Squares

A small square gives few error df, so we often run $n$ replicates (squares). The ANOVA depends on how the blocks are replicated (Montgomery, Tables 4.14–4.16): **Case 1** the same rows and columns in each replicate, **Case 2** new rows but the same columns (e.g. new batches, same operators), **Case 3** new rows and new columns.

| df | Case 1 | Case 2 | Case 3 |
|---|---|---|---|
| Treatments | $p-1$ | $p-1$ | $p-1$ |
| Rows | $p-1$ | $n(p-1)$ | $n(p-1)$ |
| Columns | $p-1$ | $p-1$ | $n(p-1)$ |
| Replicates | $n-1$ | $n-1$ | $n-1$ |
| Error | $(p-1)[n(p+1)-3]$ | $(p-1)(np-2)$ | $(p-1)[n(p-1)-1]$ |
| Total | $np^2-1$ | $np^2-1$ | $np^2-1$ |

In Cases 2 and 3 the new rows (columns) are nested in the replicates: their SS measure the variation **within** replicates.

**Crossover designs** are Latin squares in which the rows are time periods and the columns are subjects; each subject receives every treatment in turn, with a washout period against carry-over effects.

*Q:* Would you use the same Latin square in each replicate or a newly randomized one? Why?

# The Graeco-Latin Square Design

Superimpose two $p \times p$ Latin squares, one with Latin and one with Greek letters, such that **each Greek letter appears exactly once with each Latin letter**. The two squares are **orthogonal**.

**$4 \times 4$ Graeco-Latin Square Design**

| | Column | | | |
|---|---|---|---|---|
| **Row** | **1** | **2** | **3** | **4** |
| 1 | $A\alpha$ | $B\beta$ | $C\gamma$ | $D\delta$ |
| 2 | $B\delta$ | $A\gamma$ | $D\beta$ | $C\alpha$ |
| 3 | $C\beta$ | $D\alpha$ | $A\delta$ | $B\gamma$ |
| 4 | $D\gamma$ | $C\delta$ | $B\alpha$ | $A\beta$ |

- Controls three nuisance factors (rows, columns, Greek letters), or studies two treatment factors with two blocking factors, in $p^2$ runs.
- Exists for every $p \geq 3$ except $p = 6$ (and not for $p = 2$). Euler's *36 officers problem* (6 ranks $\times$ 6 regiments) has no solution; this was proved by Tarry in 1901.
- All interactions are assumed negligible. Error df: $(p-3)(p-1)$, i.e. only 4 for $p = 4$ and 8 for $p = 5$.

# Graeco-Latin Square - Rocket Propellant with Test Assemblies

## Montgomery, Example 4.4

Suppose the five test assemblies ($\alpha, \ldots, \varepsilon$) could also matter. In Montgomery's Table 4.20 the assembly in batch $i$ and operator $j$ is the Greek letter number $(i + 2j) \bmod 5$ (zero-based indices); the burning rates are the same as before.

```
greek = ["alpha", "beta", "gamma", "delta", "epsilon"]
rocket["assembly"] = [greek[(b - 1 + 2 * (o - 1)) % 5]
                      for b, o in zip(rocket.batch, rocket.op)]
gl = smf.ols("y ~ C(batch) + C(op) + C(treat)"
             " + C(assembly)", data=rocket).fit()
sm.stats.anova_lm(gl, typ=2)
               sum_sq   df         F    PR(>F)
C(batch)         68.0  4.0     2.061    0.1783
C(op)           150.0  4.0     4.545    0.0329
C(treat)        330.0  4.0    10.000    0.0033
C(assembly)      62.0  4.0     1.879    0.2076
Residual         66.0  8.0
```

Removing the assembly variability reduces $MS_E$ from 10.67 to 8.25, but the error df drop from 12 to 8, so the test is not necessarily more sensitive.

# Balanced Incomplete Block Design (BIBD)

Sometimes a block cannot hold all treatments (batch size, a day holds only $k$ runs, a car can test only some additives). In a **balanced incomplete block design**

- each block contains $k < a$ different treatments,
- every pair of treatments occurs together in exactly $\lambda$ blocks, so all pairwise comparisons are estimated with the same precision.

Parameters: $a$ treatments, $b$ blocks, $k$ treatments per block, $r$ replicates of each treatment, $N = ar = bk$ runs, and

$$\lambda = \frac{r(k-1)}{a-1} \quad \text{must be an integer;} \qquad b \geq a \text{ (Fisher's inequality).}$$

If $a = b$ the design is **symmetric**. A BIBD always exists by taking all $\binom{a}{k}$ subsets as blocks, but that is often too many blocks; smaller BIBDs are tabulated (Montgomery, Appendix; Cochran and Cox, 1957).

*Q:* Four treatments in blocks of two: how many blocks do you need? (Problem 4.45)

# BIBD - Intrablock Analysis

Model $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$ for the observed cells. Treatments and blocks are **not orthogonal**, so the treatment SS must be **adjusted for blocks**:

$$SS_T = SS_{Treatments(adjusted)} + SS_{Blocks} + SS_E, \qquad SS_{Treatments(adjusted)} = \frac{k \sum_{i=1}^{a} Q_i^2}{\lambda a},$$

$$Q_i = y_{i.} - \frac{1}{k} \sum_{j=1}^{b} n_{ij}\, y_{.j}, \qquad n_{ij} = 1 \text{ if treatment } i \text{ is in block } j,\ 0 \text{ otherwise.}$$

$Q_i$ is the total of treatment $i$ minus the average of the blocks in which it appears.

| Source | df | $F_0$ |
|---|---|---|
| Treatments (adjusted) | $a - 1$ | $MS_{Treatments(adj)} / MS_E$ |
| Blocks (unadjusted) | $b - 1$ | |
| Error | $N - a - b + 1$ | |
| Total | $N - 1$ | |

In software: put the blocks **first** in a sequential (type I) ANOVA, or use type II sums of squares. Adjusted treatment estimates: $\hat{\tau}_i = kQ_i/(\lambda a)$, $\text{Var}(\hat{\tau}_i - \hat{\tau}_j) = 2k\sigma^2/(\lambda a)$.

# BIBD - Catalyst Example

## Montgomery, Example 4.5

The reaction time of a chemical process may depend on the catalyst. Batches of raw material are blocks, but each batch is large enough for three catalysts only; the order of the catalysts within a batch is randomized.

**Balanced Incomplete Block Design for Catalyst Experiment**

| Treatment (Catalyst) | Block (Batch of Raw Material) | | | | |
|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | $y_{i.}$ |
| 1 | 73 | 74 | — | 71 | 218 |
| 2 | — | 75 | 67 | 72 | 214 |
| 3 | 73 | 75 | 68 | — | 216 |
| 4 | 75 | — | 72 | 75 | 222 |
| $y_{.j}$ | 221 | 224 | 207 | 218 | $870 = y_{..}$ |

$a = 4$ catalysts, $b = 4$ batches, $k = 3$, $r = 3$, $N = 12$, $\lambda = r(k-1)/(a-1) = 2$: each pair of catalysts appears together in two batches.

$$Q_1 = 218 - \tfrac{1}{3}(221 + 224 + 218) = -3, \quad Q_2 = -\tfrac{7}{3}, \quad Q_3 = -\tfrac{4}{3}, \quad Q_4 = \tfrac{20}{3},$$

$$SS_{Treatments(adjusted)} = \frac{3\,(9 + 49/9 + 16/9 + 400/9)}{2 \cdot 4} = 22.75.$$

# BIBD - Catalyst Example in Python

```
cat = pd.DataFrame({
    "block": [1, 1, 1, 2, 2, 2, 3, 3, 3, 4, 4, 4],
    "cat":   [1, 3, 4, 1, 2, 3, 2, 3, 4, 1, 2, 4],
    "y":  [73, 73, 75, 74, 75, 75, 67, 68, 72, 71, 72, 75]})
bibd = smf.ols("y ~ C(block) + C(cat)", data=cat).fit()
sm.stats.anova_lm(bibd, typ=1)           # blocks first
           df  sum_sq  mean_sq       F   PR(>F)
C(block)  3.0   55.00   18.333  28.205   0.0015
C(cat)    3.0   22.75    7.583  11.667   0.0107
Residual  5.0    3.25    0.650
```

The catalysts differ ($F_0 = 11.67$, $p = 0.011$). With the catalysts entered first, type I ANOVA gives the **unadjusted** catalyst SS 11.67 and the **adjusted** block SS 66.08 instead:

```
           df  sum_sq  mean_sq       F   PR(>F)
C(cat)    3.0   11.67    3.889   5.983   0.0415
C(block)  3.0   66.08   22.028  33.889   0.0010
Residual  5.0    3.25    0.650
```

**The order of terms matters** in a non-orthogonal design: always test treatments adjusted for blocks. `typ=2` gives both adjusted SS at once.

# BIBD - Adjusted Means and Tukey's Test

Compare catalysts by the **adjusted** effects $\hat{\tau}_i = kQ_i/(\lambda a)$, not by the raw means:

```
k, lam, a = 3, 2, 4
yi = cat.groupby("cat").y.sum()                 # 218 214 216 222
yb = cat.groupby("block").y.sum()               # 221 224 207 218
Q = pd.Series({i: yi[i] - yb[d.block].sum() / k
               for i, d in cat.groupby("cat")})
# -3.000 -2.333 -1.333 6.667
tau = k * Q / (lam * a)                  # -1.125 -0.875 -0.500 2.500
q = stats.studentized_range.ppf(0.95, a, bibd.df_resid) # 5.218
hsd = q * np.sqrt(k * bibd.mse_resid / (lam * a))       # 2.58
```

Adjusted means $\bar{y}_{\cdot\cdot} + \hat{\tau}_i$: 71.375, 71.625, 72.000, 75.000.

| pair | 2-1 | 3-1 | 4-1 | 3-2 | 4-2 | 4-3 |
|---|---|---|---|---|---|---|
| $\hat{\tau}_i - \hat{\tau}_j$ | 0.250 | 0.625 | 3.625 | 0.375 | 3.375 | 3.000 |
| $> 2.58$? | no | no | yes | no | yes | yes |

Catalyst 4 gives a longer reaction time than catalysts 1, 2 and 3, which do not differ. `pairwise_tukeyhsd()` would ignore the blocks and use the raw means.

# Today Exercise

**Ex04** – `code/01NAEX_Ex04.ipynb` (Python, Google Colab), a short exercise:

- The lecture examples in Python: rocket propellant (Latin and Graeco-Latin square), catalyst BIBD with adjusted means.
- Generating a randomized design and run sheet in Python (you will need it for the project).
- In-class problems from Montgomery: **4.23** (assembly methods, Latin square) and **4.40** (gasoline additives, BIBD); 4.42 as an optional extra.

There is **no separate homework** this week: the assignment is **Team Project 1**.

**Next lecture (26 October):** $2^k$ factorial designs (Montgomery, Chapter 6).

# Team Project 1 - The Pendulum Experiment

**Question:** which factors affect the period $T$ of a pendulum, and how well does the textbook formula work?

$$T_0 = 2\pi\sqrt{\frac{L}{g}} \quad \text{(small angles)}, \qquad T(\theta_0) \approx T_0\left(1 + \frac{\theta_0^2}{16} + \frac{11\,\theta_0^4}{3072}\right) \quad (\theta_0 \text{ in radians}).$$

Theory: $T$ grows with $\sqrt{L}$, does not depend on the mass, and grows slowly with the release angle:

| $\theta_0$ | $10^\circ$ | $20^\circ$ | $30^\circ$ | $45^\circ$ | $50^\circ$ | $60^\circ$ |
|---|---|---|---|---|---|---|
| $T/T_0 - 1$ | 0.19 % | 0.77 % | 1.74 % | 4.0 % | 5.0 % | 7.3 % |

For $L = 0.6\,\text{m}$ ($T_0 = 1.55\,\text{s}$) the step from $10^\circ$ to $30^\circ$ is only 0.024 s.

**Last year's lesson:** with angles of 5–20° and a stopwatch the angle effect was invisible, and *"mass is not significant"* was reported as a proof that mass does not matter.

This year you have to **plan the precision** so that the angle effect becomes detectable, and you have to show that mass does not matter with an **equivalence test**, not with a large $p$-value.

# Team Project 1 - Design Requirements

- **Factors:** length $L$ (3 levels, measured from the pivot to the centre of mass of the bob), release angle $\theta_0$ (3 levels, at least one $\leq 10^\circ$ and one $\geq 45^\circ$), mass $m$ (2 levels, the **same bob size and shape**, e.g. an empty and a filled container).
- **Response:** period $T = t_N/N$ from the time $t_N$ of $N \geq 10$ full oscillations; record the raw $t_N$.
- **Pilot study** (at least 10 timings at one setting) $\Rightarrow \hat{\sigma} \Rightarrow$ number of replicates for power $\geq 0.9$ to detect the $10^\circ$ vs $30^\circ$ difference (noncentral $F$, Lecture 3).
- **Blocking:** each replicate of the $3 \times 3 \times 2 = 18$ runs is a block (one operator or one session), as in the radar example of Lecture 3.
- **Randomization:** random run order within each block, generated in Python with a fixed seed and saved as a run sheet. If you decide not to change $L$ at random (it is laborious), say so and discuss what it means for the analysis.
- Record run order, operator, time of day and any problems (a missed count, a twisted string, . . . ).

# Team Project 1 - Analysis and Submission

**Analysis** (details in `projects/01NAEX_Project1.ipynb`):

- Factorial ANOVA in blocks for $T$ and for $\log T$; why does the $L \times \theta_0$ interaction depend on the scale?
- Multiple comparisons of the angle levels with the correct $MS_E$; relative efficiency of blocking.
- Mass: confidence interval for the mass effect and an equivalence test (TOST) with a margin you justify.
- Residual diagnostics, including residuals versus run order and operator.
- Regression of $\log T$ on $\log L$ and the angle correction $\log(1 + \theta_0^2/16 + \dots)$: estimate the exponent of $L$ (theory $1/2$) and $g$ with a confidence interval.
- Optional bonus: compare timing methods (stopwatch, video, phone sensor app) in a small Latin square with operators and sessions as blocks.

**Teams** of 2–4 students. **Submit** by pull request into `projects/`: notebook `01NAEX_Project1_<Surname1>_<Surname2>....ipynb` with outputs and the raw data as a CSV file of the same name.

**Deadline: Sunday 8 November 2026.**